Credit Risk Portfolio Analysis

PD Model v2 — Out-of-Time Validation, Loss Given Default and Underwriting Funnel

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1. Executive Summary

This report documents a probability-of-default model built on the full LendingClub loan history from 2007 to 2018 and validated out-of-time. The SQL warehouse behind it holds 2,260,668 funded loans and 27,648,741 declined applications; 1,348,099 of the funded loans have reached a final outcome and form the modelling population, of which 20.0% charged off.

The model was trained on 1,122,460 loans issued up to 2016 and tested on 225,639 loans issued in 2017 and 2018 — loans it never saw during training. On that unseen set it achieves an AUC of 0.718 and a KS statistic of 0.318. Those figures are deliberately lower than the 0.744 AUC reported by an earlier version of this model, which used a random split inside a single 2015 vintage. A random split lets loans originated weeks apart, under identical economic conditions and underwriting policy, sit on opposite sides of the split; it flatters the result. The out-of-time figures are the ones that describe how the model would behave in production.

Across the 225,639 scored loans, carrying $3.26B of exposure, average predicted default probability is 20.4% against an actual realised default rate of 21.3%. Applying a 90% loss given default — measured from recovery data rather than assumed — gives an expected loss of $662.6M against $693.2M actually incurred, so the model under-predicted losses by 4.4%.

Two findings carry beyond the model itself. First, recoveries on charged-off loans return only 10.0% to 11.6% of the outstanding balance, and that figure barely moves across risk grades — so loss given default sits near 90% portfolio-wide, not the 100% placeholder an earlier version of this analysis assumed. Second, the declined-application file supports an underwriting view the funded data cannot provide on its own, and analysing it surfaced a data quality defect severe enough to distort any model that used the field naively (Section 6).

2. Data and Methodology

2.1 Data sources

- Funded loans: 2,260,668 records, 2007–2018, loaded into a SQL warehouse with origination terms, borrower credit bureau attributes as at application, and full payment history.
- Declined applications: 27,648,741 records over the same period, carrying amount requested, debt-to-income, state, employment length and a risk score.
- Modelling population: 1,348,099 funded loans with a resolved outcome. Loans still Current, Late or In Grace Period are excluded entirely — their final outcome is unknown, and treating them as non-defaults would bias the default rate downward.
- Target: default = 1 for Charged Off or Default, 0 for Fully Paid. Overall rate 20.0%.

2.2 Leakage control

Credit models are unusually prone to target leakage because the richest fields in a loan file describe what happened after origination. Every such field is excluded here: principal and interest received, recoveries, outstanding balance, last payment date and amount, updated FICO scores, and the hardship and debt-

settlement flags. Only information available at the moment of approval is used as a feature. This matters for honesty as much as rigour — several published analyses of this dataset report AUCs above 0.90, which is attainable only by including fields that encode the answer.

A second, subtler form of leakage was also controlled for. LendingClub introduced a number of credit bureau fields partway through the data history, so for older vintages they are empty. 14 features more than half empty across the training window were dropped, because retaining them would let the model learn "is this field populated" as a proxy for how recently a loan was issued.

2.3 Validation design

Training covers loans issued up to 2016 (1,122,460 loans, 19.7% default rate); testing covers 2017 and 2018 only (225,639 loans, 21.3%). This is the standard applied in regulated credit model validation, and it is strictly harder than a random split: the test period carries its own economic conditions and its own underwriting policy, neither of which the model saw.

HistGradientBoostingClassifier was selected over a logistic regression baseline, which reached an AUC of 0.706 on the same split. 176 features were used after one-hot encoding.

3. Portfolio Findings

3.1 Risk grading

LendingClub assigns each loan a grade from A to G at origination. Observed outcomes confirm the grading is meaningful: the default rate rises monotonically across all seven grades, and the dollar loss rate rises faster still, because higher-grade loans are also larger and priced higher.

Grade	Loans	Funded	Wtd Avg Rate	Default Rate	Loss Rate
A	235,193	$3.26B	7.11%	6.0%	3.2%
B	393,102	$5.20B	10.66%	13.4%	7.8%
C	382,323	$5.42B	14.07%	22.4%	14.6%
D	201,657	$3.07B	17.76%	30.4%	21.0%
E	94,192	$1.65B	21.14%	38.4%	27.5%
F	32,306	$613.9M	24.92%	45.1%	33.8%
G	9,326	$190.1M	27.61%	49.7%	37.7%

All 1,348,099 resolved loans. Loss rate is net dollar loss as a share of funded principal.

3.2 Loan purpose

Purpose remains informative after grade is accounted for. Small business lending is the clear outlier on the riskier end, while car loans and credit card refinancing sit at the lower end.

Purpose	Default Rate	Loans
Small Business	36.1%	2,368
Moving	26.2%	1,994
Medical	23.6%	3,742
Debt Consolidation	22.3%	123,790
Other	22.3%	18,135
Major Purchase	21.7%	6,097
House	20.7%	2,279
Vacation	19.9%	2,239
Home Improvement	18.4%	18,273
Credit Card	18.4%	43,673

Out-of-time test set, purposes with at least 200 loans.

3.3 Recovery and loss given default

When a loan charges off, the lender recovers only a small fraction of the outstanding balance through collections — and strikingly, that fraction is almost independent of the borrower's risk grade. A grade G default returns about as little as a grade A default.

Grade	Defaults	Recovery Rate	Implied LGD
A	14,214	10.0%	90.0%

Grade	Defaults	Recovery Rate	Implied LGD
B	52,661	10.4%	89.6%
C	85,805	10.7%	89.3%
D	61,264	10.8%	89.2%
E	36,199	11.3%	88.7%
F	14,585	11.6%	88.4%
G	4,632	11.3%	88.7%

Recovery rate is total recoveries divided by unpaid principal at charge-off.

This flatness is what justifies applying a single portfolio-wide loss given default of 90% in the expected-loss calculations throughout this report. An earlier version of this analysis used 100% as a deliberately conservative placeholder, before the recovery data had been examined.

3.4 Vintage trend

Default rates move with the credit cycle. The 2008 and 2009 vintages carry elevated rates, conditions improve through 2010 to 2013, and rates climb again from 2014 onward as origination volume grew.

Issue Year	Resolved Loans	Default Rate
2007	603	26.2%
2008	2,393	20.7%
2009	5,281	13.7%
2010	12,537	14.0%
2011	21,721	15.2%
2012	53,367	16.2%
2013	134,804	15.6%
2014	223,103	18.4%
2015	375,546	20.2%
2016	293,105	23.3%
2017	169,321	23.1%
2018	56,318	15.8%

3.5 Geographic concentration

Exposure concentration and risk concentration are not the same map. California, Texas and New York carry the largest balances, but New York defaults materially above California at comparable scale.

State	Loans	Exposure	Default Rate
CA	196,853	$2.89B	19.6%
TX	110,354	$1.67B	19.8%
NY	110,097	$1.57B	22.1%
FL	95,843	$1.31B	21.5%

State	Loans	Exposure	Default Rate
IL	51,870	$762.7M	18.1%
NJ	48,583	$735.8M	21.1%
PA	45,656	$639.9M	20.8%
GA	43,480	$633.6M	18.4%

4. Model Performance

4.1 Out-of-time versus random split

The table below compares this model against the earlier version built on a random split within a single vintage. The newer model trains on seventeen times more data and still scores lower, which is the expected and correct result.

Metric	v2 (out-of-time)	v1 (random split)
AUC-ROC	0.718	0.744
KS statistic	0.318	0.360
Brier score	0.151	0.139
Training loans	1,122,460	65,918
Test loans	225,639	21,973

Both generations used the same algorithm and the same leakage controls. Only the validation design differs.

4.2 Discrimination and calibration

Ranking the test set by predicted probability and splitting it into ten equal groups shows both whether the model separates risk and whether its probabilities are honest. Predicted and actual rates track closely in every decile, which is calibration — a more demanding property than discrimination, and the one that matters when the output feeds a loss provision rather than just a ranking.

Decile	Loans	Predicted PD	Actual Rate	Gap	Lift
10	22,564	49.4%	47.3%	2.1 pp	2.22x
9	22,564	35.3%	35.8%	-0.5 pp	1.68x
8	22,564	28.0%	29.9%	-1.9 pp	1.41x
7	22,564	23.1%	24.9%	-1.8 pp	1.17x
6	22,563	19.3%	21.1%	-1.8 pp	0.99x
5	22,564	16.0%	17.7%	-1.6 pp	0.83x
4	22,564	13.0%	14.0%	-1.1 pp	0.66x
3	22,564	10.0%	11.1%	-1.1 pp	0.52x
2	22,564	6.9%	7.5%	-0.7 pp	0.35x
1	22,564	3.5%	3.6%	-0.1 pp	0.17x

Decile 10 is the highest predicted risk. Lift is the decile default rate relative to the portfolio average.

4.3 Feature importance

Permutation importance, measured as the drop in AUC when a feature is randomly shuffled, identifies which inputs the model actually relies on.

Rank	Feature	Importance
1	int_rate	0.0494
2	term_months	0.0121
3	loan_amnt	0.0096

Rank	Feature	Importance
4	dti	0.0067
5	annual_inc	0.0049
6	fico_range_low	0.0029
7	installment	0.0027
8	mort_acc	0.0026
9	grade_B	0.0019
10	mo_sin_old_rev_tl_op	0.0018
11	emp_length_years	0.0017
12	home_ownership_RENT	0.0017

Interest rate dominates because it encodes the lender's own risk assessment at origination — it is a legitimate origination-time field, but it means the model partly learns to reproduce an existing pricing decision.

5. Risk Segmentation and Expected Loss

Scored loans are grouped into six bands by predicted probability. The bands behave as intended: predicted and actual default rates rise together, and the two highest bands carry 21.6% of the loans.

Risk Band	Loans	% of Book	Exposure	Predicted PD	Actual Rate	Expected Loss
1 Low	27,816	12.3%	$362.7M	3.9%	4.1%	$12.7M
2 Low-Medium	43,741	19.4%	$531.3M	9.0%	9.9%	$43.3M
3 Medium	57,303	25.4%	$745.1M	15.9%	17.4%	$107.0M
4 Medium-High	47,964	21.3%	$723.8M	24.5%	26.4%	$160.5M
5 High	34,012	15.1%	$603.4M	36.4%	36.6%	$198.6M
6 Very High	14,803	6.6%	$293.2M	53.1%	50.6%	$140.5M

Expected loss is predicted PD × funded amount × 90% LGD, summed per loan.

Aggregate expected loss of $662.6M compares with $693.2M actually realised — an under-prediction of 4.4%. For a provisioning application this direction of error matters more than its size: a model that systematically under-states losses produces reserves that are too thin, and would warrant a conservatism adjustment before use.

5.1 Watchlist

The fifteen highest-risk loans in the scored book are listed below, with what actually happened to each. Of the top twenty by predicted probability, 15 went on to default.

Loan ID	Grade	Amount	Purpose	State	Predicted PD	Outcome
111757961	G3	$15.9K	Debt Consolidation	NY	81.3%	Defaulted
95886454	G5	$32.0K	Small Business	CA	80.8%	Repaid
111008192	G4	$30.0K	Other	CA	80.0%	Defaulted
111976854	D5	$25.0K	Debt Consolidation	OK	79.9%	Repaid
102648623	F1	$25.0K	Debt Consolidation	TX	79.1%	Defaulted

Loan ID	Grade	Amount	Purpose	State	Predicted PD	Outcome
134871757	E5	$30.0K	Debt Consolidation	PA	78.8%	Defaulted
109786394	F1	$30.0K	Debt Consolidation	GA	78.7%	Defaulted
110430372	E1	$24.0K	Debt Consolidation	AR	78.5%	Defaulted
125033188	G1	$35.0K	Debt Consolidation	TX	78.2%	Defaulted
118162507	G1	$35.0K	Debt Consolidation	KY	78.0%	Defaulted
111095805	D5	$18.0K	Debt Consolidation	TX	77.4%	Defaulted
113499400	G5	$28.0K	Debt Consolidation	CA	77.4%	Repaid
125028022	E5	$21.5K	Other	NC	77.3%	Defaulted
97330426	F5	$15.2K	Debt Consolidation	NY	77.3%	Defaulted
131915438	F1	$20.0K	Home Improvement	NY	77.1%	Defaulted

Every loan here is from the out-of-time test set, so the outcome column is a genuine test of the ranking rather than a restatement of training data.

6. Underwriting Funnel and Reject Inference

The funded loan file describes only applications that were approved, which means it cannot answer who gets declined or why. The 27,648,741 declined applications close that gap. An acceptance model built on the fields common to both populations — amount requested, debt-to-income, application year and state — reaches an AUC of 0.672, rising to 0.855 when an approximate credit score is included, with the score the dominant coefficient. That ordering is consistent with how underwriting actually operates.

6.1 A data quality finding

An initial version of this model included employment length and scored an AUC of 0.908, which was implausibly strong for four weak inputs. The cause was a defect in the data rather than a signal in it: employment length reads "less than one year" for 83% of declined applications against 8% of funded ones. That is a form default on applications declined before employment details are collected, not a real difference in borrower tenure. The field was excluded, and the lower figure is the one reported.

This is worth recording as a methodological point rather than a footnote. A model that is unexpectedly accurate deserves suspicion before celebration, and the usual explanation is that a field encodes the outcome rather than predicting it.

7. Recommendations

- Provision using the measured 90% loss given default rather than a flat assumption. Recoveries are consistently minimal and do not vary meaningfully by grade, so a single portfolio-wide figure is defensible and materially more accurate than a 100% placeholder.
- Apply a conservatism adjustment when using model output for provisioning, given the model under-predicts realised losses on unseen data.
- Validate out-of-time before any deployment decision. The gap between the two validation designs documented here is the difference between a model that looks deployable and one that is.
- Review pricing on grades E through G, where default rates reach 49.7% and loss rates exceed a third of funded principal.
- Monitor state-level risk separately from state-level exposure, since the two do not move together.
- Treat the declined-application file as an analytical asset for underwriting review, with attention to fields that carry form defaults rather than observed values.

8. Limitations and Assumptions

- Loss given default is measured at the point of charge-off. It does not discount recoveries received in later periods, and takes no account of the time value of money. A full LGD model would do both.
- The model contains no macroeconomic variables. It describes how default rates moved across vintages without explaining the credit cycle driving them, which limits its usefulness for forward-looking stress scenarios.
- The credit score used in the acceptance model compares the risk score recorded on declined applications against the FICO range recorded on funded ones. These are similar measures but are not confirmed to share a definition or a scale, so that model is indicative rather than precise.
- Interest rate is the strongest single predictor, which means the model partly reproduces the lender's own pricing decision rather than forming an independent view of risk.

- Full reject inference techniques, such as fuzzy augmentation or a Heckman-style correction, were not applied. The acceptance model stands alongside the PD model rather than feeding into it.
- Results describe a single lender's book over one eleven-year period, ending in 2018. Transfer to another portfolio or a later period should not be assumed.